

ARATHON CHALLENGE 03

Data-driven rural infrastructure placement

Where would one additional relay, repeater, small cell, or measurement campaign deliver the greatest improvement?

Research rationale

Why it matters A hackathon result becomes operationally valuable when it moves from describing weak service to recommending a limited, defensible intervention.

Testable hypothesis Optimizing placement against predicted performance deficits and route importance will outperform choosing only the single worst measured point.

Suggested technical approach

Method Use the digital-twin surface to define underserved demand points, generate feasible candidate locations, and solve a constrained facility-location problem. Include an alternative objective that chooses new measurement points to reduce model uncertainty.

Primary inputs Predicted service surfaces, uncertainty, GPS route density, base-station distance, radio quality, performance metrics, and optional terrain or siting constraints.

How a team can demonstrate success

Evaluation Compare before-and-after population or route coverage under explicit service thresholds; test robustness to model uncertainty; report gains per intervention and sensitivity to placement constraints.

Demo artifact A scenario planner where judges place one or more assets and immediately see estimated coverage, performance, uncertainty, and route-benefit changes.

Scope note Use geographically separated test segments so nearby samples do not appear in both training and evaluation data.